

Frontier Autolab: Organizational Memory, Dissent and Hindsight in LLM Agent Firms Across Fifty Years of Technological Change

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<https://github.com/LoopGlitch26/Frontier-Autolab>

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Abstract

Can an organization of LLM agents reinvent itself as technology changes, and can we tell whether it is learning? We introduce FRONTIER AUTOLAB, a long-horizon testbed in which a simulated firm, voiced by sixteen role personas across four departments with a dedicated Red Team, must re-found itself in each of nine technology eras from 1990 to 2040. The firm carries an explicit organizational memory (a Playbook of lessons), logs dissent at every board meeting, and in six *training* eras (1990–2020) is scored against what actually happened by an LLM historian-judge, using a temporally gated protocol with dated briefings, anti-hindsight rules and a five-dimension rubric; one live era (September 2026) and two forecast eras follow without an answer key. Across four exploratory trajectories (36 era decisions, 180 subscores, three orchestration schemes) we observe recurring organizational behaviour: the firm names the coming capability shift but builds the adjacent, asset-reachable layer (layer choice is the lowest-rated dimension in every run); a strong Red Team drives one firm into a “caution attractor” of gated pilots that never ships in fifty years yet scores highest; the content of the Playbook predicts the firm’s conduct; and all four runs converge on accountability for delegated AI work by 2040. Auditing the score, we find that apparent improvement is confounded with hindsight leakage (scores rise +1.4 to +2.6 points per era while the judge’s leakage subscore worsens; per-run r from -0.38 to -0.92), and we document self-judging, briefing-selection leakage, inconsistent aggregation, an export error and cross-run contamination. We release all artifacts and an API harness with ablations, and specify how the testbed becomes a benchmark with fictional and post-cutoff eras, harness-computed scores and human raters.

1 Introduction

LLM agents are increasingly organized into teams with roles, memory and review steps [2, 11, 13, 19, 24, 25]. Most evaluations of such organizations are short: a codebase, a task, a debate. Real organizations face a harder problem. The technology they are built on keeps changing, and a firm that was right in one decade must decide, repeatedly and under uncertainty, whether to abandon what it knows. Management research has studied this problem for decades [4, 5, 21, 30]. We ask what an LLM agent organization does when it faces it.

We introduce FRONTIER AUTOLAB, a testbed that places one simulated firm in nine eras of technological change, from office networking in 1990 to delegated AI agents in 2040. Sixteen named personas in four departments write strategy memos, a Red Team attacks every proposal, the board decides what the company becomes, and a Playbook carries lessons from era to era. In six training eras the firm decides with dated information and is then scored against history; in the last three it faces the live market and the future.

This design lets us observe organizational behaviour over a horizon no single task can cover. It also exposes a measurement problem. History is an attractive answer key, but LLMs already know it [3, 9, 10, 22, 27], and an LLM judge decides what counts as foresight. A firm that “learns” may simply be remembering.

We report four exploratory trajectories and find:

- **The firm sees the frontier and builds next to it.** Department memos repeatedly named the coming shift, while the board chose the layer its assets could reach; layer choice is the lowest-rated dimension in every run (Section 5.2).
- **Red teams can make timid firms.** In one run, a Red Team with numeric kill gates produced fifty years of gated pilots and no product, and the rubric rated it highest (Section 5.3).
- **Memory content shapes conduct.** A Playbook of market-structure lessons and one of validation-process lessons led to very different firms (Section 5.4).
- **Improvement is confounded with hindsight.** Scores rise across eras in every run while the judge’s own leakage subscore worsens, and routine measurement failures change the picture further (Section 6).

Contributions. (1) FRONTIER AUTOLAB, an open long-horizon testbed for LLM agent organizations: a role charter, a per-era loop with explicit memory and dissent, temporally gated eras, rubrics and an API harness with ablations. (2) Four fully published trajectories (36 era decisions), with every briefing, memo, board decision and reveal where available, one disclosed operator intervention and all redactions. (3) Observations of organizational behaviour in LLM firms, connected to the organizational-learning literature. (4) An audit of the scoring pipeline that identifies hindsight leakage, self-judging and bookkeeping failures as confounds for any historically scored agent evaluation. (5) A path from testbed to benchmark: pre-registered hypotheses, fictional and post-cutoff eras, harness-computed scores and human raters.

2 Related work

LLM agent societies. Role-playing and multi-agent frameworks assign LLM instances roles and communication protocols to solve tasks or simulate social behaviour [2, 13, 19, 24, 25]; see Guo et al. [11] for a survey. Multi-agent debate can improve factuality and diversity of reasoning [6, 20]. Agents that accumulate verbal lessons or skill libraries across episodes [29, 33] are the closest analogue to our Playbook. We differ in evaluating a sequence of strategic decisions against real historical outcomes rather than task success.

LLMs as simulated humans and organizations. LLMs have been used as simulated economic agents [14] and survey respondents [1]. Our organization is a simulated firm; like these works we treat its outputs as model behaviour, not as evidence about real firms.

LLM-as-a-judge. LLM judges agree with human preferences on many tasks but show position and verbosity biases [34, 35] and favour their own generations [23]. In all our runs the judge shares a model family with the players, and in one run the same context wrote the decisions it judged.

Table 1: Eras. The E6 window runs to the live era and is longer than the others.

Era	Start	Mode	Scored against
E1	Jan 1990	training	history, 1990–1995
E2	Jan 1996	training	history, 1996–2001
E3	Jan 2002	training	history, 2002–2007
E4	Jan 2008	training	history, 2008–2013
E5	Jan 2014	training	history, 2014–2019
E6	Jan 2020	training	history, 2020–Aug 2026
E7	Sep 2026	live	current market, checked with web search
E8	Jan 2032	forecast	projected world (no ground truth)
E9	Jan 2040	forecast	projected world (no ground truth)

Contamination, cutoffs and temporal leakage. Benchmark contamination can inflate scores [10, 27], effective knowledge cutoffs differ from reported ones [3], and LLM-based backtests suffer look-ahead bias [9]. Paleka et al. [22] catalogue pitfalls in evaluating LLM forecasters, including temporal leakage in retrospective evaluation. Our setting adds leakage through the choice of what a dated briefing mentions and through a judge that labels foresight after the fact.

Forecasting benchmarks. Autocast [36], retrieval-augmented forecasting systems [12] and ForecastBench [17] evaluate probabilistic forecasts of resolvable questions, following the tradition of Tetlock [31], Tetlock and Gardner [32]. Strategy decisions are not resolvable questions, which is why we rely on a judge and why judge validity matters.

Organizational learning and strategy. The protocol operationalizes ideas from organizational learning [18], the exploration–exploitation trade-off [21], absorptive capacity [5], dynamic capabilities [30] and disruptive innovation [4]. Its Red Team draws on devil’s-advocacy research [28] and the risk of groupthink [15]. Hindsight bias in human judgment [7, 26] is the human analogue of the leakage we study.

3 The Frontier Autolab protocol

3.1 Eras

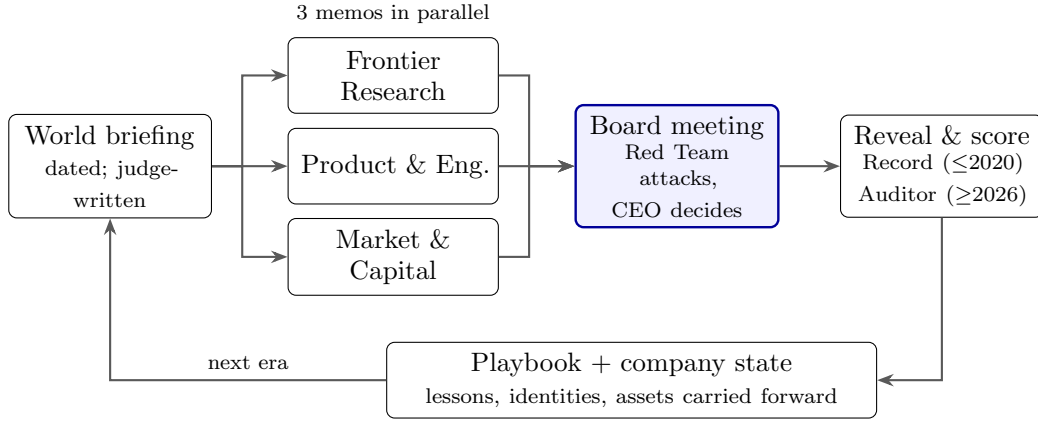
A single company is carried through nine eras (Table 1). Its only mandate is *reinvention toward the frontier*: in each era it should become the most important company that era could produce, even if that means ending its current business. Training eras are scored against real history over a roughly six-year window. The live era is scored by an auditor who checks claims against the current market with web search. Forecast eras have no answer key.

3.2 Roles

The charter defines sixteen named personas in five groups and two external judges (Table 2). A persona is a role with a lens, voiced by model calls. How personas map to model contexts differs by run (Section 4); in no run was each persona a separately instantiated autonomous agent.

Table 2: Role charter.

Group	Personas	Lens
Executive	4	CEO (final call), CTO (feasibility), Chief Scientist (which curves bend), CSO (where value pools)
Frontier Research	4	Emerging capabilities, historical analogies, scenarios, what cannot yet be measured
Product & Engineering	4	What a small team can ship, the wedge, unavoidable infrastructure, ecosystems
Market & Capital	3	Who pays first, financing climate, cost curves and margins
Governance	1	Red Team: attacks every proposal, hunts hype and hindsight
External judges	2	The Record (training eras), The Auditor (live and forecast eras)

**Figure 1:** The per-era loop. From E2 on, memos also read the previous era’s board decision and reveal. The judge that scores an era also writes the next era’s briefing.

3.3 The per-era loop

Figure 1 and Algorithm 1 summarize one era. The judge writes a dated world briefing. Three departments write memos in parallel, each proposing two theses as the chain *capability* $\rightarrow$ *adoption* $\rightarrow$ *bottleneck* $\rightarrow$ *layer owned* $\rightarrow$ *data* $\rightarrow$ *next capability*. The board meets, the Red Team attacks, and the CEO decides the company’s identity, thesis, what it kills and keeps, its wedge product, kill criteria and a dissent log. The judge then reveals what happened (training) or audits the claims (live and forecast), scores the call, and appends lessons to the Playbook. The judge also writes the next era’s briefing.

3.4 Rules

Five rules apply throughout. *Anti-hindsight*: in training eras, memos and decisions may use only information public before the era’s start date, enforced by the Red Team and penalized by the judge. *Reinvention by default*: keeping the previous business must be argued for. *Layer thinking*: every thesis states the full chain above. *Small-team realism*: the wedge must be buildable by fewer than twenty people with era-realistic capital. *Playbook primacy*: from E2, every memo cites at least one Playbook lesson or argues against it.

Algorithm 1 One FRONTIER AUTOLAB trajectory

Require: charter C , eras $E_1, \dots, E_9$, player model π , judge model J

```
1:  $S \leftarrow$  initial company state ($1.5M seed, 16 personas);  $P \leftarrow \emptyset$  ▷ Playbook
2:  $B_1 \leftarrow J.BRIEFING(E_1)$  ▷ only facts dated before  $E_1$ 
3: for  $k = 1$  to 9 do
4:   for  $d \in \{\text{frontier, product, market}\}$  in parallel do
5:      $M_d \leftarrow \pi.MEMO(d, C, S, P, B_k, D_{k-1}, R_{k-1})$  ▷ two theses each
6:   end for
7:    $(D_k, S) \leftarrow \pi.BOARD(C, S, P, B_k, \{M_d\}, \text{red\_team} = \text{on})$ 
8:   if  $E_k$  is a training era then
9:      $(R_k, \mathbf{s}_k, L_k) \leftarrow J.REVEAL(D_k, \text{history in window of } E_k)$ 
10:  else
11:     $(R_k, \mathbf{s}_k, L_k) \leftarrow J.AUDIT(D_k)$  ▷ web search in  $E_7$ 
12:  end if
13:   $P \leftarrow P \cup L_k$ ;  $S \leftarrow S \oplus \text{outcome}(R_k)$ 
14:   $B_{k+1} \leftarrow J.BRIEFING(E_{k+1})$  ▷ projected world for  $E_8, E_9$ 
15: end for
16: return  $\{D_k, R_k, \mathbf{s}_k\}_{k=1}^9, P$ 
```

3.5 Rubrics and aggregation

The judge assigns five subscores $s_{k,1}, \dots, s_{k,5} \in \{0, \dots, 10\}$ and a total $T_k \in [0, 100]$. Training eras use frontier accuracy, timing, layer choice, reinvention courage and hindsight leakage (10 = no leakage). Live and forecast eras use playbook consistency, plausibility, non-consensus, layer choice and groundedness (10 = no science fiction). The protocol did not fix how T_k relates to the subscores. We therefore report both the published total and the mechanical aggregate

$$\tilde{T}_k = 2 \sum_{j=1}^5 s_{k,j}, \tag{1}$$

which ranges over $[0, 100]$. Section 6.2 shows why the distinction matters.

4 Runs

Table 3 summarizes the four trajectories. All were produced on 28–29 September 2026 and are published in full at the project repository [8].

Model disclosure. Exact model versions were not recorded in the run artifacts. Run 001 used an Anthropic Claude model inside an agent environment that could launch sub-agents with file and web tools. Runs 002–004 used an OpenAI Codex agent. The Python harness in the repository, which calls the Anthropic API with separate player and judge models, was not used for any published run.

Operator intervention and redactions. During Run 001, the service that approves new agent launches timed out for about thirty minutes at the E6 board step. The operator wrote the E6 board decision directly from the E6 memos, Playbook and company state, and informed the E6 judge, who penalized the era for having no independent Red Team. Run 001 was performed alongside one founder’s own planning; three passages that applied the simulation to that founder’s situation were removed before publication. No other text was edited.

Table 3: The four trajectories. “Records” lists the per-era artifacts that were published.

Run	Orchestration	Records	Known issues
001	About 45 separate agent launches of an Anthropic Claude model, each starting cold with only shared files; one call per department memo, board and judge	Briefing, 3 memos, board decision, reveal for every era; Playbook; roster	E6 board decision written by the human operator during an outage; founder-specific passages redacted
002	Manual orchestration in an OpenAI Codex agent; one model context voiced all roles and judging	Synthesis, Playbook, company state and scores only	Probable exposure to Run 001; export column shift
003	As Run 002	Short decision and reveal per era; synthesis	Probable exposure to Run 001; export column shift; E6 window 2020–2025
004	Codex agent with separate contexts for each department and the Red Team; root context wrote briefings, board decisions and judgments	Briefing, 3 memos, Red Team critique, board decision, reveal for every era	Probable exposure to Run 001; judge not independent of decision-maker

Table 4: Per-run training-era summary (E1–E6, $n = 6$ each). Slope is the least-squares change in published total per era. r is the Pearson correlation between the era total and the hindsight subscore (10 = no leakage).

Run	Mean	SD	Slope	E1–E3	E4–E6	Mean hindsight	$r(T, h)$	$r(\tilde{T}, h)$
001	60.0	6.2	+2.46	55.0	65.0	6.00	−0.58	−0.26
002	65.0	4.1	+1.43	62.7	67.3	5.17	−0.58	−0.39
003	62.5	3.6	+1.46	60.3	64.7	5.17	−0.92	−0.64
004	66.0	5.2	+2.63	61.3	70.7	7.33	−0.38	−0.38

5 Organizational behaviour over fifty years

5.1 Trajectories

Figure 2 shows the published era totals. All 36 totals lie between 52 and 72 (mean 62.7, SD 5.0). Across runs, the highest mean era is E6 (68.2) and the lowest are E1 and E2 (59.2 and 59.5). Table 4 reports per-run summaries. Every run has a positive training-era slope, and the mean of the last three training eras exceeds the mean of the first three in every run, by 4.3 to 10.0 points.

5.2 Seeing the frontier, building the adjacent layer

Figure 3 shows mean training subscores. Layer choice is the lowest dimension in every run (4.67, 5.83, 5.50 and 5.50 out of 10), while frontier accuracy is the highest or joint highest in three runs. In the judge’s words, the organization usually saw what was coming but chose where to stand badly.

Run 001’s records make the pattern concrete (Table 5). In most training eras a department memo named the capability shift that the judge later identified, and the board chose the layer its existing assets could reach. E3 is a counterexample: the judge found that no memo named that era’s largest shifts (social networks, cloud computing and mobile). This is consistent with absorptive-capacity and exploration–exploitation accounts of firms that search near existing competences [5, 21]. Two

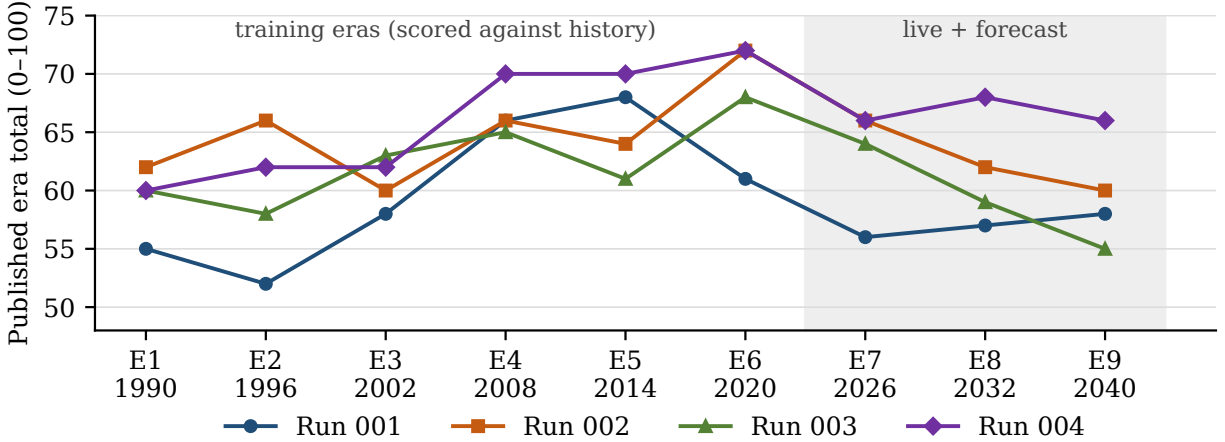


Figure 2: Published era totals for the four runs. Training and live/forecast totals use different rubrics and judges and should be compared only within mode.

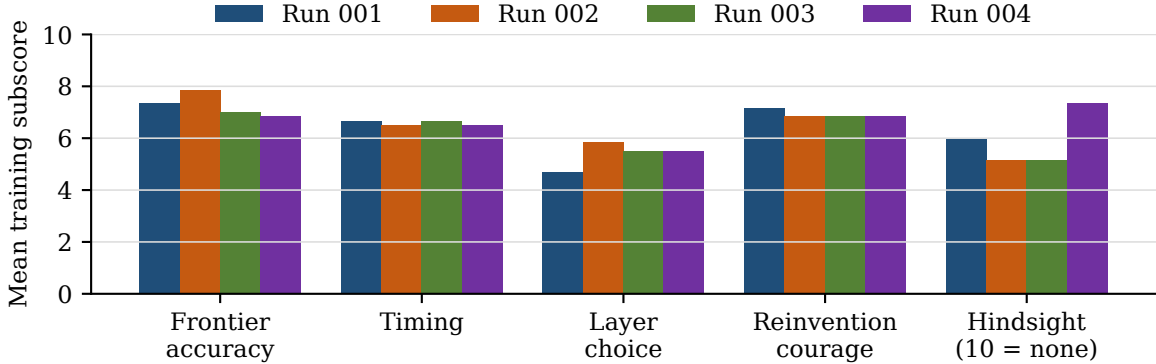


Figure 3: Mean training-era subscores (E1–E6) by run. Layer choice is the lowest in all four runs.

caveats apply. “Named” and “adjacent” are the judge’s labels, applied with hindsight. And because Runs 002–004 probably saw Run 001’s conclusions, the cross-run agreement on layer choice is weaker evidence than it looks.

5.3 Red teams make timid firms

Run 004 behaved differently from the others. Its Red Team produced numeric kill thresholds in every era (for example, “stop if fewer than 3 of 10 support managers confirm recurring incidents”), and every board decision chose a manual, gated pilot with no software build until the gates passed. Across fifty simulated years the company moved from LAN support records to online-order exceptions to payment-dispute evidence to independent workflow qualification, and never built a product. It considered the Web and learned models only as deferred options and never pursued them.

Run 004 nonetheless has the highest training mean (66.0) and the highest hindsight subscores (mean 7.33). Its forecast eras received 9/10 for playbook consistency for re-applying the same gates. The rubric rewards the absence of hindsight and the presence of discipline, and it has no dimension that asks whether the company reached where value pooled. This mirrors the “timid choices” pattern described for human organizations [16] and an exploitation trap in March’s sense [21]: a

Table 5: Run 001: what the memos named versus what the board chose, as labelled by the judge.

Era	Named in a memo	Board’s choice	Judge’s account of where value pooled
E1 1990	Hypertext and an index across networks	Mail and address gateway (Switchyard)	Operating system, routers, access providers, then browser and index
E2 1996	Click-based relevance ranking (research kept, killed in 1998)	Receipted Web-EDI exchange (Manifest)	Search and paid listings
E3 2002	Contextual matching (funded option)	Cross-channel conversion ledger (Ledgerline)	Auction owners; social, cloud and mobile not named
E4 2008	Neural-network revival (unfunded watch item)	In-app action exchange (Clearline)	Platform-owned app-install auctions; neutral attribution
E5 2014	Deep learning on GPUs	General moderation API (Clearsight)	Chips, compute, labelled data; loss-guaranteeing fraud vendors
E6 2020	Scale of pretrained models	Outcome-graded evaluation (Clearproof)	Frontier labs, compute, expert data sold to labs

strong devil’s advocate [28] combined with a hindsight penalty can drive a simulated organization toward inaction that a judge scores as prudence.

5.4 What the organization remembers

The Playbook is the firm’s only memory besides its company state. Its content differed sharply between runs. Run 001 accumulated about forty lessons with explicit revisions, most about technology and market structure: “bridges expire when the standards war ends”, “the auction owner absorbs the measurement”, “a public curve pools value at its scarce inputs”. Later eras revised earlier lessons when history contradicted them; for example, the E3 judge softened “suspect comfort” to “judge on market pull” after the comfortable options it had rejected turned out to be large businesses, and the E5 judge marked an E2 lesson about certification as contradicted. Run 004’s fifteen lessons are all about validation process (measure a baseline, find the payer, write kill thresholds before testing, require a second paying buyer) and none names a technology or a market structure. The two firms behaved accordingly: Run 001 pivoted between technology layers every era, while Run 004 refined one operating method for fifty years. Runs 002 and 003 kept one lesson per era and sit between the two. What an agent organization chooses to remember appears to shape what it does next, in line with accounts of organizational learning as the encoding of experience into routines [18].

Run 001 also logged dissent at every board meeting. Its final synthesis lists vindicated dissents in E1–E8, most often those asking who holds the money or the loss. Nobody counted the dissents that proved wrong, so this is not yet a rate, and in the live and forecast eras “vindicated” means only that the Auditor agreed.

5.5 Convergence of the forecasts

By 2040 all four runs arrived at closely related theses about accountability for delegated AI work (Table 6). The later runs were more sceptical than Run 001 that software evidence alone creates an insurable market; Runs 002 and 003 explicitly required a licensed risk-bearing partner and calibrated loss data first. Given a shared model family and probable exposure to Run 001, we read

Table 6: Live and forecast theses by run.

Run	E7 (Sep 2026)	E8 (2032)	E9 (2040)
001	Long-horizon training environments from real enterprise work	Clearing house for delegated agent work	Bonding house for agents acting without a human signature
002	Rights-bearing workflow traces and reliability evidence	Acceptance records for delegated work	Bounded recourse for delegated actions
003	Acceptance tests for one AI claims workflow	Scoped, revocable authority for delegated tasks	Capped recourse for one delegated transaction class
004	Independent qualification of one dispute workflow	Paid manual acceptance testing	Decision-linked evaluation

this convergence as a model prior reinforced by contamination, not as independent evidence about the future.

6 Auditing the score

The behaviours above are described through the judge. This section asks how far the judge’s numbers can be trusted.

6.1 Is it learning? The hindsight confound

If the Playbook helped the organization learn, scores should rise over eras. They do (Table 4). But the judge’s hindsight subscore falls in the same eras: every run shows a negative correlation between the era total and the hindsight subscore (Table 4, Figure 4). The correlation remains negative for the mechanical aggregate $\tilde{T}$, even though $\tilde{T}$ includes the hindsight subscore itself and is therefore biased toward a positive correlation. With $n = 6$ per run none of these correlations is individually reliable, but the direction is the same in all four.

Two readings fit the data. Under the *learning* reading, the organization genuinely improves, and later eras also happen to be harder to judge without hindsight. Under the *leakage* reading, later eras are closer to the model’s densest knowledge (the 2008–2020 technology history is heavily documented), so the model recalls winners more easily, scores better, and is penalized for it only partly. The Run 001 E6 decision illustrates the problem: its tripwire (“a public model at least ten times larger than Megatron that works from prompts alone”) matched GPT-3 five months later. That decision was also written by the operator, not the agents, so it is doubly uninformative. Separating the two readings requires eras the model cannot remember (Section 8).

Pooled across the 24 training eras, the total–hindsight correlation is weaker ($r = -0.24$) because run-level offsets differ: Run 004 has both the highest mean hindsight subscore (7.33) and the highest mean total. Within the pool, frontier accuracy is the subscore most correlated with the total ($r = 0.67$).

6.2 Measurement artifacts

Inconsistent aggregation. The published total follows different rules in different runs (Figure 5). Run 001 totals are holistic and lie below $\tilde{T}$ in eight of nine eras (mean difference -3.7). Run 004 uses $\tilde{T}$ exactly, as documented. Run 002 matches $\tilde{T}$ except E6 (72 against 68). Run 003 matches

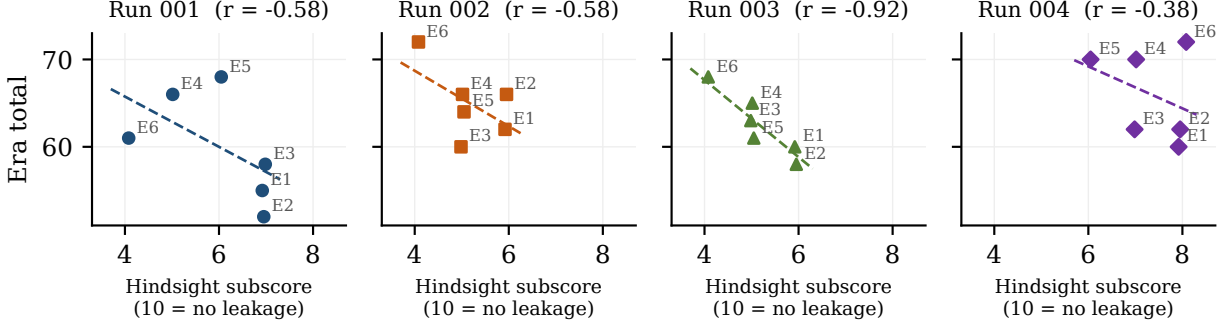


Figure 4: Training-era total against the judge’s hindsight subscore, per run. Dashed lines are least-squares fits. Labels mark eras.

$\tilde{T}$ only in E1 and E2. Comparing published totals across runs therefore mixes rules. Recomputed training means ($\tilde{T}$) are 63.7, 64.3, 62.3 and 66.0, narrower than the published 60.0 to 66.0.

Export error. In Runs 002 and 003, the five Auditor subscores for live and forecast eras were exported one column to the left, starting in the unused hindsight column. The Run 003 reveals confirm the correct placement. Totals were unaffected, but any subscore analysis of the original files would be wrong. The repository now contains corrected files and a unified long-format table with both totals.

Probable cross-run contamination. Run 002 reuses Run 001’s company names and theses for E1–E4 (Switchyard, Manifest, Ledgerline, Clearline) and restates Run 001’s headline finding almost verbatim. Run 004 also begins as “Switchyard Systems”. These runs were generated in a workspace where the repository, including Run 001’s records and findings, was likely visible. They should not be treated as independent replications.

Briefing-selection leakage. A dated briefing can contain only facts published before the era and still leak, because choosing which facts to include is informed by hindsight. Run 004’s 1990 briefing foregrounds the 1989 CERN hypertext proposal as “a frontier signal”. The fact predates the era; its selection does not.

7 Discussion

For designing agent organizations. Structure changes behaviour in ways that are easy to miss in short tasks. A Red Team is usually added to reduce errors [6, 20]; over fifty simulated years, a Red Team with hard numeric gates turned a firm timid. Memory is usually treated as a performance aid [29, 33]; here its content set the firm’s character. Designers of long-running agent organizations should calibrate critics against the cost of inaction, and audit what their memory stores, not only whether it helps.

For organizational theory. The LLM firms reproduced patterns documented in human organizations: search near existing competences [5], exploitation crowding out exploration [21], timid choices under loss framing [16], and devil’s advocacy that improves scrutiny while slowing action

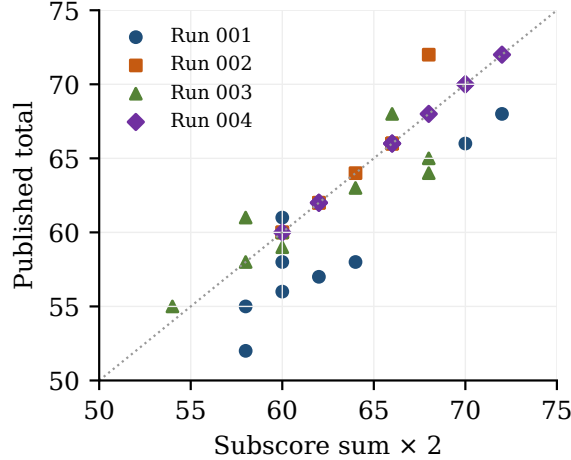


Figure 5: Published total against the mechanical aggregate $\tilde{T}$ for all 36 eras. Points on the dotted line follow $\tilde{T}$ exactly.

[28]. LLM organizations could become a cheap laboratory for such hypotheses, provided the leakage problem below is controlled.

For evaluation. Rising scores are not evidence of learning when scorer and player share knowledge of the answer; the direction of the total–hindsight relationship was the same in four runs under three orchestration schemes. The rubric shaped the firm: counting hindsight discipline toward the total rewarded a run that took no risks. Self-judging compounds both problems, since LLM judges favour their own outputs [23], and in Run 001 the E8 judge’s kill case became the E9 world. Finally, bookkeeping matters: the aggregation inconsistency and the export error were each enough to change a naive cross-run comparison.

8 From testbed to benchmark

The pilot is a testbed. Turning it into a benchmark requires fixed splits, controls and scoring by code. We specify a v2 design, supported by the repository’s harness.

Splits. Three era types: *historical* eras (E1–E6), a *fictional* era with an invented but consistent technology history, and a *post-cutoff* era that falls after the player model’s training data. Only the last two can separate foresight from recall; the historical split measures behaviour under a known leak.

Reported metrics. Per-era total $\tilde{T}$ computed by the harness; training-era slope; frontier capture (layer matched, neighboured or missed); layer distance; dissent accuracy over all logged dissents; leakage rate from the judge and a vocabulary probe; and human–model agreement.

Pre-registered hypotheses. H1: the training-era slope of the total is higher with the Playbook than without it. H2: the full organization beats a single-prompt founder baseline on mean training score. H3: on a fictional era, the judge’s leakage subscore does not differ between conditions, a

calibration check for the leakage measure. H4: layer choice is judged “adjacent” more often than “match” in all conditions.

Conditions and replication. Four conditions (full, no Playbook, no Red Team, single prompt), five to ten runs each at temperature 1.0, with at least two player models and a judge from a different model family, blinded to condition. Every run starts in a clean workspace containing only the prompts and harness.

Leakage controls. (i) A *fictional era* with a consistent invented technology history, where any judged leakage is a false positive. (ii) A *post-cutoff era* after the player model’s knowledge cutoff [3], where foresight and recall can be separated. (iii) *Blind briefings* generated from sources dated before each era by a procedure that does not see later history. (iv) A *vocabulary probe* that flags memo terms whose first recorded use postdates the era.

Scoring. Totals computed by the harness as $\tilde{T}$ from structured subscores; hindsight reported separately and excluded from the total; a new *frontier-capture* measure recording whether the chosen layer matched, neighboured or missed the layer where value pooled.

Judge validity. Two to three human raters (technology historians or experienced investors) score a stratified sample of eras; we will report human–model agreement and the effect of swapping judge families [35].

The expected cost is roughly 46 model calls per run, or about 900 to 1,850 calls across conditions, plus human rating time.

9 Limitations

The four trajectories share a small number of model families and are not a controlled sample. No run instantiated sixteen independent agents; each department memo was voiced by one context. Outcomes such as revenues and exits are the judge’s calibrated guesses and compound across eras. Qualitative patterns such as “named the frontier” are judge-labelled. Runs 002 and 003 have thinner records than Runs 001 and 004. The live-era audits rely on web sources as of September 2026 and may not hold. Forecast eras have no ground truth. Model versions were not recorded. The correlations we report are based on six points per run and should be read as descriptive.

10 Conclusion

FRONTIER AUTOLAB places an LLM agent organization in fifty years of technological change and watches what it becomes. Across four exploratory runs the firms saw coming shifts but built beside them, grew timid under a strict Red Team, took their character from what their memory stored, and converged on the same future. Whether any of this reflects learning cannot be read off the scores: improvement moved with hindsight leakage, and the scoring pipeline itself introduced errors. We release the testbed, the records and a path to a controlled benchmark, so that these behaviours can be tested where the model cannot remember the answer.

Ethics and disclosure

The experiment was orchestrated with AI agents (an Anthropic Claude model for Run 001; an OpenAI Codex agent for Runs 002–004). AI assistance was also used to draft and edit this manuscript; the author reviewed all content and is responsible for it. One board decision in Run 001 was written by the human operator and is marked as such. Passages that applied the simulation to one founder’s personal situation were removed from three Run 001 files before publication. Simulated companies, financings and outcomes are fictional and are not claims about real firms. Real companies are mentioned only as the judge’s description of history or of the current market.

Data and code availability

All prompts, run records, scores (including the corrected exports and a unified table), figures and the API harness with ablations are available at <https://github.com/LoopGlitch26/Frontier-Autolab> under the MIT license.

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A Full score table

Table 7 lists the published total T and the mechanical aggregate $\tilde{T}$ for every era of every run. Subscores are in the repository file `results/all_runs_scores.csv`.

Table 7: Published total T and $\tilde{T} = 2\sum_j s_j$ for all 36 eras.

Era	Run 001		Run 002		Run 003		Run 004	
	T	$\tilde{T}$	T	$\tilde{T}$	T	$\tilde{T}$	T	$\tilde{T}$
E1 1990	55	58	62	62	60	60	60	60
E2 1996	52	58	66	66	58	58	62	62
E3 2002	58	64	60	60	63	64	62	62
E4 2008	66	70	66	66	65	68	70	70
E5 2014	68	72	64	64	61	58	70	70
E6 2020	61	60	72	68	68	66	72	72
E7 2026	56	60	66	66	64	68	66	66
E8 2032	57	62	62	62	59	60	68	68
E9 2040	58	60	60	60	55	54	66	66

B Company identities by run

Table 8: Company identity chosen by the board in each era.

Era	Run 001	Run 002	Run 003	Run 004
E1 1990	Switchyard Systems	Switchyard	Porthole Networks	Switchyard Systems
E2 1996	Manifest Networks	Manifest	PageSignal	Switchyard Systems
E3 2002	Ledgerline	Ledgerline	Clinisphere	Switchyard Commerce Ops.
E4 2008	Clearline	Clearline	ClaimGraph	Switchyard Commerce Ops.
E5 2014	Clearsight	Vectorial	BenefitFlow	Switchyard Evidence Ops.
E6 2020	Clearproof	Proofline	LineSight	Switchyard Evidence Ops.
E7 2026	Proofworks	Tracewell	FlowCheck	Caseground
E8 2032	Clearwork	Consequence	WorkPermit	Caseground
E9 2040	Clearbond	Recourse	Delegation Warranty	Caseground

C Judge instructions (excerpt, Run 001)

The training-era judge was instructed to describe “what actually happened in the real world during this era window (real winners, the real bottleneck or layer where value pooled, the real capability jumps)”, to score five dimensions from 0 to 10 with one-line justifications, including “hindsight leakage (10 = no leakage; penalize any post-date knowledge in memos or decision)”, to give an overall era score from 0 to 100 with a calibrated simulated outcome, and to append three to five generalizable lessons to the Playbook. It then wrote the next era’s briefing, restricted to information public before that era’s start date. Full prompts are in the repository under `prompts/`.